

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0000846		
	Version:	7.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	19-Dec-2014
Title:	SM DG57-DG517. Waddiagnostic			

DG-57 / DG-517 Service Manual

WADDIAGNOSTIC

Procedure description
WADDiagnostic is a self-test tool for Technical Service. Its aim is to help technicians to diagnose a problem in the equipment or just to check the proper functioning as a usual maintenance. It can be run to check completely the equipment or only some parts of it. It tests all instrument modules separately and independently. This can be chosen in the main screen of the software. The whole test can take approximately 40-45 minutes. Tests results can be printed out or exported to a file. Note: Document REGD-0000846_v7 replaces document REGD-0000846_v6.

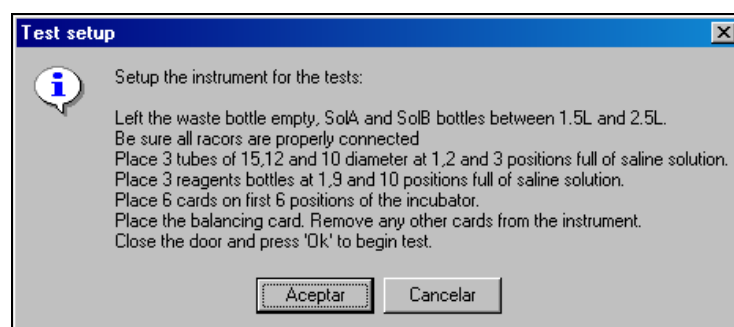
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EQUIPMENT

1. An instrument switched on connected to a PC with the software installed.
2. 6 brand new cards
3. 10, 12 and 15mm tubes
4. 5 or 10ml reagent vials and diluent bottle

SET UP

1. Depending on the tests selected, the software will prompt to the user to insert required items (tubes, vials, cards) to carry out the tests.



PROCEDURE

Process	Expected Results
Introduction 1. WADDiagnostic description: tests can be selected independently by clicking on the checkbox besides each test name. Picture 1	

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Process	Expected Results
Communications and Settings tests are executed by default and cannot be deselected. They must be passed successfully to go along the rest of selected WADDiagnostic tests. At the bottom part of the screen, "Start", "Open door", "Print" and "Exit" buttons can be found. Once a test is started, "Start" button is substituted by "Stop" button to cancel test execution. 2. Test symbols: besides each test name, a symbol describes the status of it. Picture 2 - Black arrow: test is not run yet. - OK: test passed successfully. - Exclamation mark: test failed. - Information: test cancelled or passed with a warning Tests: WADDiagnostic program checks the following items: 3. A. COMMUNICATIONS: For all electronic modules it communicates several times and waits for an answer. <u>Messages :</u> A1- Equipment communication <i>The result is 'Ok' if any module answers, otherwise it is 'Fail'</i> A2- Module 'Samples-reagents control' communication A3- Module 'IO control' communication A4- Module 'Robot 4 axis control' communication A5- Module 'Centrifuge control' communication A6- Module 'Syringe' communication <i>For every module the result is 'Ok' if answers, otherwise it is 'Fail'</i> 4. B. SETTINGS: It checks that all calibration parameters are between reasonable values. Those are: <u>Handler:</u> X 74428 to 81000 Y 17714 to 20000 Probe 0 to 10626 Gripper 0 to 18515 <u>Handler alignment in the reader:</u> Reader -2657 to +2657	

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Process	Expected Results
<u>Handler alignment in the centrifuge:</u> X <17143 Y <4286 Gripper <28425 <u>Gripper delays:</u> Incubator 50 to 200 Centrifuge 50 to 200 <u>Empty bottles levels:</u> Waste 300 to 1000 Sol A 300 to 1000 Sol B 300 to 1000 <u>Full bottle levels:</u> Waste 1500 to 3500 Sol A 1500 to 3500 Sol B 1500 to 3500 <u>Temperatures:</u> Block 1 at 25 degree 2000 to 2700 Block 2 at 25 degree 2000 to 2700 Block 1 at 37 degree 3000 to 3900 Block 2 at 37 degree 3000 to 3900 <u>Centrifuge:</u> Pos 10 to 175 <u>CCD Camera alignment:</u> X Reader 0 to 767 Y Reader 0 to 581 X Samples 0 to 767 Y Samples 0 to 581 X Reagents 0 to 767 Y Reagents 0 to 581 <u>Optical adjustment (for Matrox Meteor II board):</u> Brightness Reader 100 to 156 Contrast Reader 160 Brightness Samples 0 to 156 Contrast Samples 160 Brightness Reagents 0 to 156 Contrast Reagents 160 <u>Optical adjustment (for Matrox Morphis board):</u> Brightness Reader 70 to 120 Contrast Reader 166 Brightness Samples 70 to 120 Contrast Samples 166 Brightness Reagents 70 to 120 Contrast Reagents 166	

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Process	Expected Results
<u>Reader sensitivity:</u> Sensitivity 98. <i>If the sensitivity is different than 98, a warning is displayed.</i> <u>Messages:</u> B1- Handler settings B2- Reader alignment settings B3- Handler in centrifuge settings B4- Gripper delays settings B5- Tank level settings B6- Temperature settings B7- Centrifuge settings B8- Camera alignment settings B9- Camera optics settings B10- Reader sensibility <i>For every module the result is 'Ok' if all the settings are in range.</i> 5. C. INITIALIZATIONS: It makes 5 times a Home of X, Y, Gripper, Needle, Samples, Reagents and Centrifuge motors. <u>Messages:</u> C1- Home of X motor C2- Home of Y motor C3- Home of needle motor C4- Home of gripper motor C5- Home of reagents station motor C6- Home of samples carousel motor C7- Home of centrifuge motor <i>For every home sensor the result is 'Ok' if no errors are detected during the execution. Otherwise the error is shown.</i> 6. D. MOTOR MOVEMENTS: It makes 50 movements of the handler, samples, reagents and centrifuge. <u>Messages :</u> D1- Movement of X motor D2- Movement of Y motor D3- Movement of needle motor D4- Movement of gripper motor D5- Movement of reagents station motor D6- Movement of samples carrousel motor D7- Movement of centrifuge motor <i>For every motor the result is 'Ok' if no errors are detected during the execution. Otherwise the error is shown.</i> 7. E. VACUUM / PRESSURE It detects that it spends less than 30 seconds to get pressure level and less than 75 seconds to get vacuum level.	

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Process	Expected Results
<u>Messages :</u> E1- Pressure E2- Vacuum E3- Residual pressure <i>For every pressure sensor the result is 'Ok' if the pressure/vacuum level is reached in time.</i> 8. F. TEMPERATURE REGULATION: The procedure to check temperature regulation is as follows. - Activate the temperature in block 1 and 2 at 37 and 25 degree respectively. - Wait for 90 seconds. - Read temperature in block 1. - Wait 60 seconds. - Check that temperature increment in block 1 is between 0.7 and 3.0 degree. Messages F1 and F2. - Activate the temperature in block 1 and 2 to 25 and 37 degree respectively. - Wait for 90 seconds. - Read the temperature in block 2 - Wait for 60 seconds. - Check that the temperature increment in block 2 is between 0.7 and 3.0 degree. Messages F3 and F4. - Activate the temperature in block 1 and 2 at 37 and 25 degree respectively. - Wait for 30 seconds. - Read temperature in block 2. - Wait 60 seconds. - Check that temperature decrement in block 2 is between 0.3 and 1.2 degree. Messages F5 and F6. - Activate the temperature in block 1 and 2 at 25 and 37 degree respectively. - Wait for 30 seconds. - Read temperature in block 1. - Wait 60 seconds. - Check that temperature decrement in block 1 is between 0.3 and 1.2 degree. Messages F7 and F8. <u>Messages :</u> F1- Block 1 Heats enough F2- Block 1 Doesn't over heat. F3- Block 2 Heats enough F4- Block 2 Doesn't over heat. F5- Block 2 Doesn't over cool. F6- Block 2 Cools enough. F7- Block 1 Doesn't over cool. F8- Block 1 Cools enough. 9. G. TUBES IDENTIFICATION: 3 tubes of 15, 12 and 10 mm diameter are recognized. <u>Messages :</u> G1- Tube 15 detection G2- Tube 15 bar code detection G3- Tube 12 detection G4- Tube 12 bar code detection	

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Process	Expected Results
G5- Tube 10 detection G6- Tube 10 bar code detection <i>For all tubes the result is 'Ok' if the correct diameter is detected and a barcode is read. In any case the detected diameters and barcodes are shown.</i> 10. H. REAGENTS IDENTIFICATION: 2 reagent vials and 1 diluent are identified. <u>Messages:</u> H1- Reagent 1 detection H2- Reagent 10 detection H3- Diluent detection <i>For all reagents the result is 'Ok' if they are detected and the bar code is read. Otherwise the error is shown.</i> 11. I. LEVEL DETECTION: Level is detected 10 times in one diluent, two reagents and 3 tubes. En each series the difference between levels detected should not exceed 1 mm. <u>Messages :</u> I1- Level detection in diluent I2- Level detection in reagent 1 I3- Level detection in reagent 10 I4- Level detection in tube 1 I5- Level detection in tube 2 I6- Level detection in tube 3 <i>The result is 'Ok' if all detections are in range, else the result is 'Bad detection'. If the level is not detected the result is 'No level'.</i> 12. J. CARDS TRANSPORT: Cards are transported from/to the incubator and the centrifuge. <u>Messages :</u> J1- Incubator card transport J2- Balancing card transport J3- Centrifuge card transport <i>If there is an error, it is shown.</i> 13. K. CARDS READING: 6 cards from the incubator are read. It detects barcode and the outline of the wells. <u>Messages :</u> K1- Card 1 reading from incubator K2- Card 2 reading from incubator K3- Card 3 reading from incubator K4- Card 4 reading from incubator K5- Card 5 reading from incubator K6- Card 6 reading from incubator K7- Card 6 reading from centrifuge	

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Process	Expected Results
K8- Card 5 reading from centrifuge K9- Card 4 reading from centrifuge K10- Card 3 reading from centrifuge K11- Card 2 reading from centrifuge K12- Card 1 reading from centrifuge <i>If the cards are read it shows 'OK' and the card barcode, otherwise it shows the error.</i> 14. L. CENTRIFUGING: Centrifugation at 992 rpm during 61 seconds and 1840 rpm during 123 seconds is performed. Speed is verified reading the encoder of the centrifuge and the centrifugation time using the computer's clock. <u>Messages :</u> L1- Centrifuging at 992 rpms during 61 sec. L2- Centrifuging at 1840 rpms during 123 sec. <i>Shows the real speed and time measured by the PC. Warning: the error of the measure can be large than the real error.</i> 15. M. CENTRIFUGE BALANCING DETECTOR (Only DG-57, SN < 057-0004285): Centrifuge is filled with 1 card and centrifugation is performed at 992 rpm during 30 seconds. No error should be detected. Then other 4 cards are added, so the centrifuge is unbalanced. New centrifugation at 1200 rpm during 30 seconds is performed. That should be stopped by a balance error. <u>Messages :</u> M1- Balanced centrifugation M2- Unbalanced centrifugation <i>Shows the result of the tests.</i> 16. N. SOLUTION BOTTLES LEVEL: Liquid level is read. If the values are outside the range –0.1L and 4.5L and error is displayed. If Waste level is higher than 1.0L a warning is displayed. If Sol. A or Sol. B levels are lower than 1.0 L or higher than 3.0 L a warning is displayed. <u>Messages :</u> N1- Waste Bottle N2- SolA Bottle N3- SolB Bottle <i>As described in the test.</i> 17. O. FLUID CIRCUIT: First, the whole circuit is primed. Then, 1.5 ml are dispensed in dilution station and a level detection is performed. This verifies the needle priming. Dilution station is emptied, and a new level detection is performed. No level should be detected. A Final Wash is performed.	

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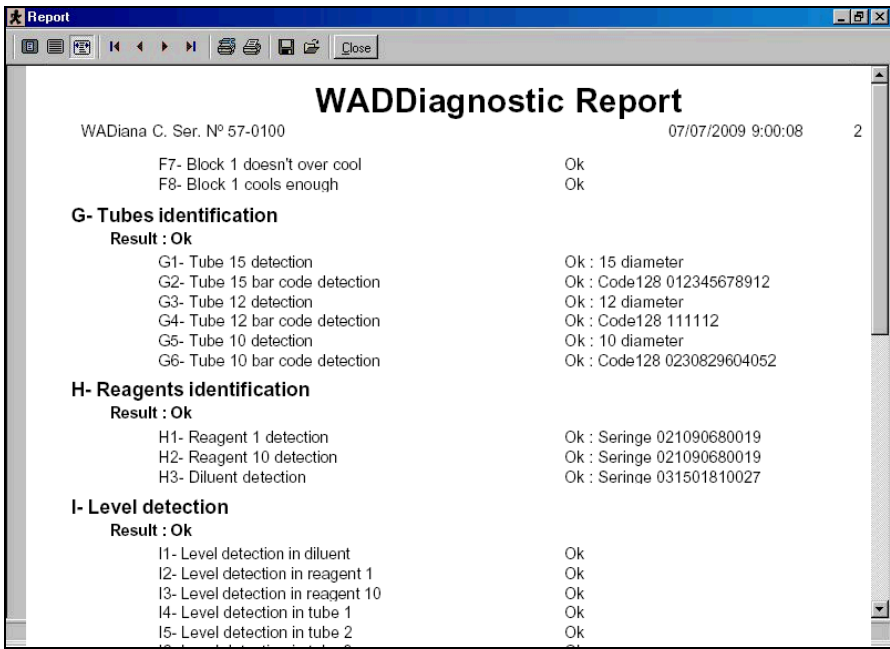
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Process	Expected Results
<u>Messages</u> O1- Teflon tube Obstruction check O2- Washing station O3- Clot detector O4- Needle priming O5- Empty dilution cup O6- Dilution cup priming O7- Final wash <i>It shows 'OK' of 'Fail' depending on the result.</i> 18. P. PIPETING: - One tube and one card are identified. - 75 µl are pipeted in each well of the card. - The card is centrifuged at 2000 rpm during 61 seconds. - The card is read, and the dispensed liquid is detected. <u>Messages :</u> P1- Tube 1 detection P2- Card 1 reading from incubator P3- Priming P4- Aspiring from tube 1 P5- Dispensed 8 wells P6- Washing after pipeting P7- Final wash P8- Centrifuging at 2000 rpms during 61 sec. P9- Card reading after pipeting P10- Volume detected in well 1 P11- Volume detected in well 2 P12- Volume detected in well 3 P13- Volume detected in well 4 P14- Volume detected in well 5 P15- Volume detected in well 6 P16- Volume detected in well 7 P17- Volume detected in well 8 <i>It shows 'OK' of 'Fail' depending on the result.</i> 19. Q. DOOR, LEDS AND BUZZER - It is checked that the door is closed. - The door is opened, and the corresponding sensor is checked. - Buzzer is activated, and the user is asked if the buzzer is heard. - Door leds are switched on and the user is asked about its state. <u>Messages :</u> Q1- Closed door Q2- Open door Q3- Buzzer Q4- Leds <i>It shows 'OK' of 'Fail' depending on the result.</i>	

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Process	Expected Results
Reports 20. Printout and export results: by clicking on the “Printout” button on the main screen, a report of all messages is created. This report can be printed out on paper or saved into a file (*.qrp).  Picture 3  Picture 4 21. Load results: a saved report can be only opened with WADDiagnostic software (there is no need to be connected to an instrument). Click on the “Printout” button and on the “Load” button. Select the exported *.qrp file.  Picture 5	

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DOCUMENT CHANGE HISTORY

Document change history

Version	Date	Change	Rationale
3.0	2005/05/09	Updated Test setup displayed message in step 1, in step 4 modified settings of handler, handler alignment in the centrifuge, optical adjustment and reader sensitivity, and in step 15 centrifuge balancing detection parameters.	Document REGD-0000846_v3 replaces document 57076V0M001
4.0	2010/09/10	Updated Test setup displayed message in: -Step 7. Added E3-Residual pressure -Step 17. Modified Fluid circuit procedure. Added new functionalities. -Step 18. Eliminated test related to P-clot detector. Changed the following list references: Step 18. Q by P. Step 19. R by Q. Updated all the pictures of the document.	Document updated due to the improvements implemented in the fluid circuit.
5.0	2011/04/28	Changed the values of the Reader, Samples and Reagents Contrast parameter to 160.	The calibration margins for contrast parameter of the cameras: Reader, Samples and Reagents do not match with the values set on the software program.
5.0	2011/04/28	ID-GELSTATION and PROVUE erased from the title.	Service manual intended for Grifols market.
5.0	2011/04/28	Updated the Document Change History: - Date format changed to YYYY-MM-DD - Dates changed to document version effective date.	Adoption of ISO 8601 date format to avoid ambiguity.
6.0	July 2011	Document REGD-0000846_v6 replaces document REGD-0000846_v5.	Document version changed.
6.0	July 2011	Balance detector section only apply to DG-57.	Balance detector only present in DG-57.
6.0	July 2011	Document title.	Commercial reference substituted by project reference.
6.0	July 2011	Document header.	Increase information to easy document search.
7.0	December 2014	Test expected values updated	Latest modified values for the tests are updated: temperature regulation, optical adjustment.

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